

Eavesdrop

⚠ **Security Warning:** Work in an isolated environment (virtual machine or container). Never open unknown files on your primary machine.

Challenge Description

Download this packet capture and find the flag.

Hints

- All we know is that this packet capture includes a chat conversation and a file transfer.
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Tools Used

- `file` – Identify PCAP file type
 - **Wireshark** – GUI tool for analyzing network traffic (TCP stream following)
 - `tshark` – Command-line Wireshark tool for packet extraction
 - `sort` – Organize timestamps in chronological order
 - `cut` – Extract specific fields from output
 - `xxd` – Convert hexadecimal to binary data
 - `openssl` – Decrypt DES3-encrypted files
 - `cat` – Read file contents
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Complete Solution

Step 1: Download and Prepare

Download the PCAP file named `capture.flag.pcap` from the challenge page.

Step 2: Basic File Inspection

First, verify the file type and general information:

```
file capture.flag.pcap
```

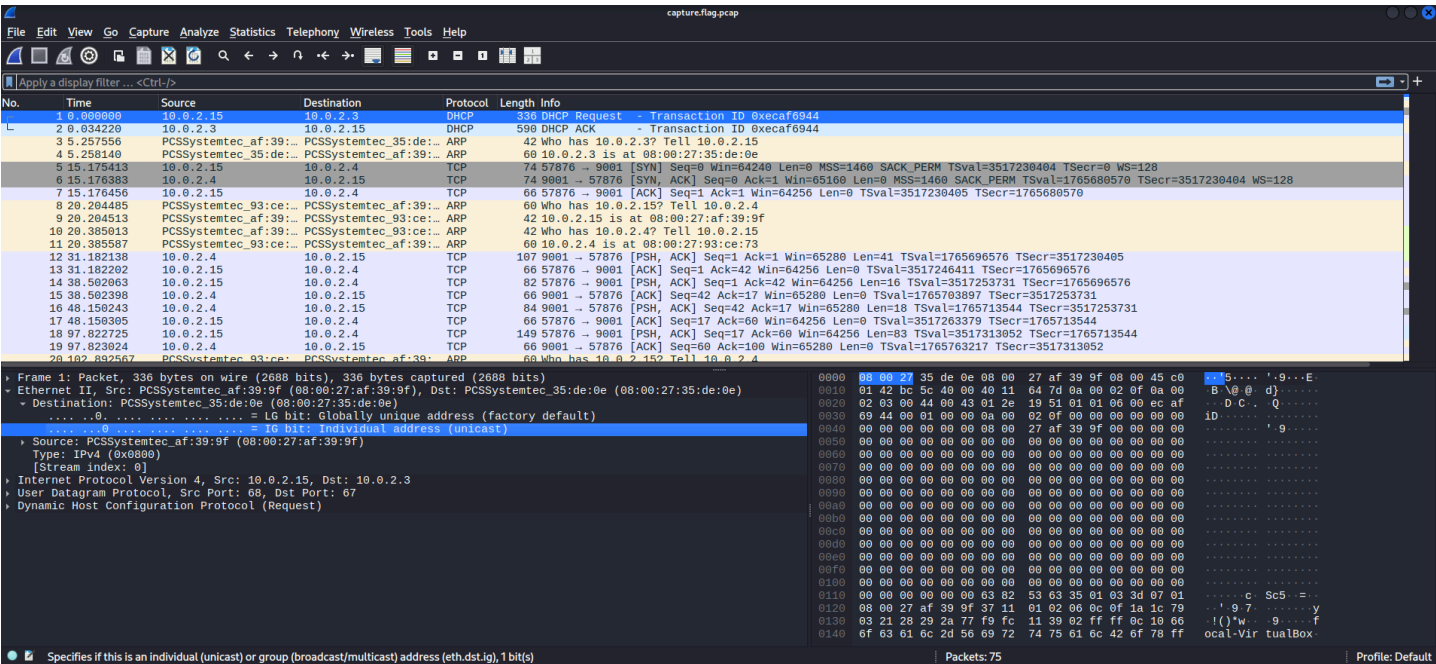
Expected output:

```
capture.flag.pcap: pcap capture file, microsecond ts (little-endian) - version 2.4
(Ethernet, capture length 262144)
```

The file is a network packet capture (PCAP) that requires analysis using network forensics tools.

Step 3: Initial Analysis with Wireshark

Open the PCAP file in Wireshark to get a visual understanding of the traffic:

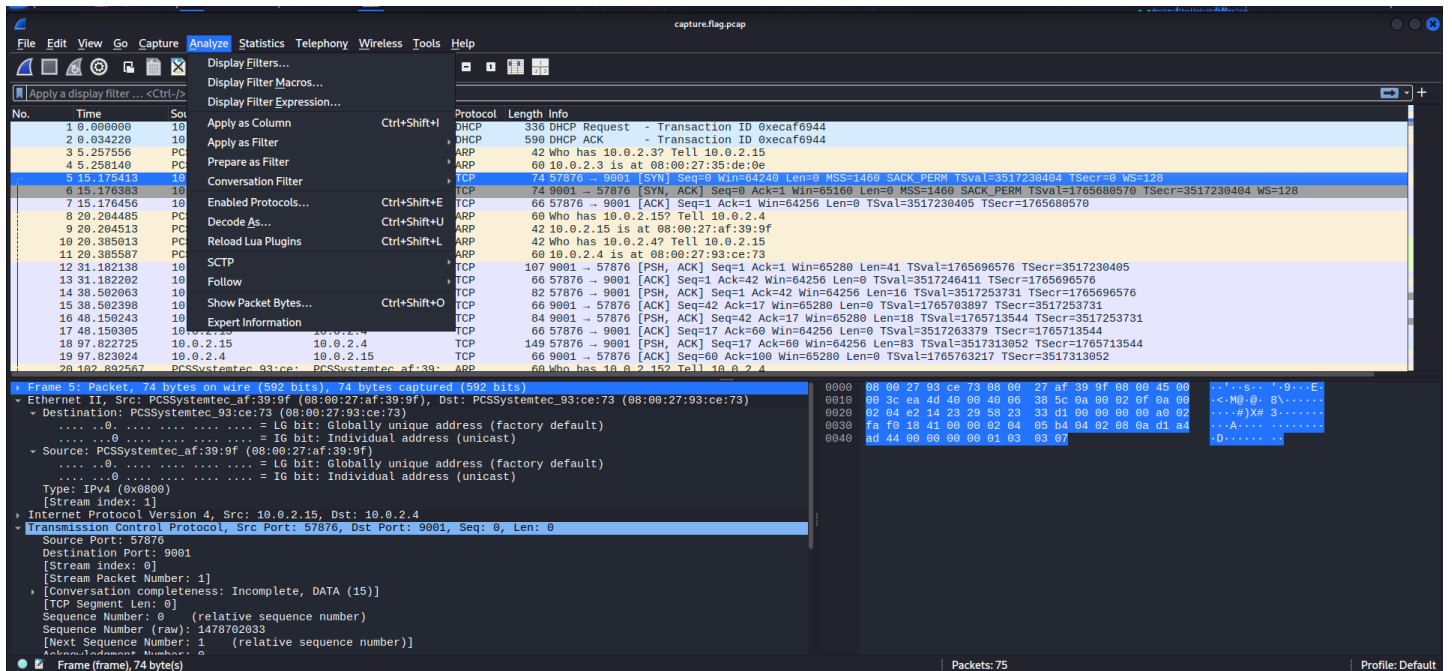


The hint mentions a **chat conversation**. In network traffic, chat conversations typically occur over **TCP**.

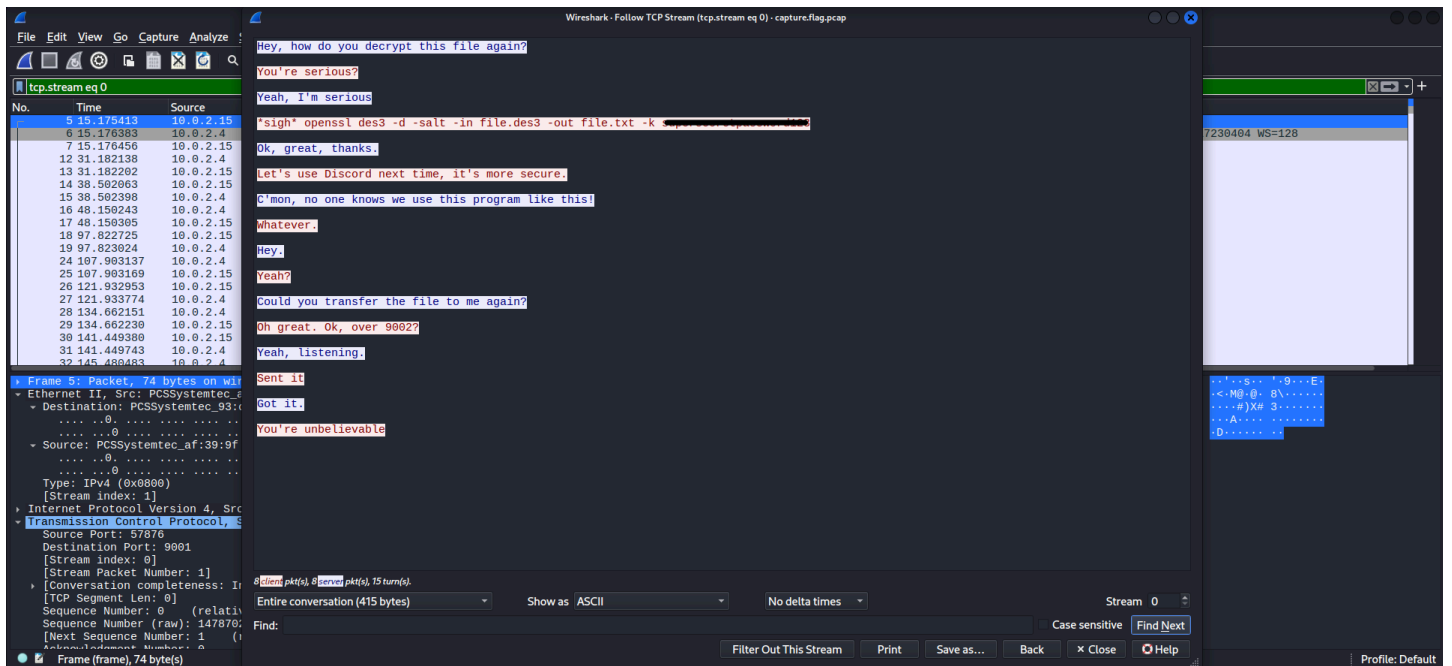
Step 4: Follow the TCP Stream

To examine the chat conversation:

1. Select any TCP packet in the capture
2. Go to **Analyze** → **Follow** → **TCP Stream**



The TCP stream reveals the chat conversation:



Extracted conversation:

* * *

Could you transfer the file to me again?
 Oh great. Ok, over 9002?
 Yeah, listening.
 Sent it
 Got it.

You're unbelievable
* * *

Key findings:

- A file transfer occurred over **port 9002**
- The command `openssl des3 -d -salt -in file.des3 -out file.txt -k <REDACTED>` was mentioned

Step 5: Extract the Encrypted File (Port 9002)

Now we need to extract the file that was transferred over port 9002. Let's use `tshark` to capture all payload data:

```
tshark -r capture.flag.pcap -Y "tcp.port==9002" -T fields -e frame.time -e tcp.payload
```

Output:

```
2021-10-04T20:09:31.525353000+0200
2021-10-04T20:09:31.526250000+0200
2021-10-04T20:09:31.526326000+0200
2021-10-04T20:09:31.526588000+0200
53616c7465645f5f3c4b26e8b8f91e2c4af8031cfaf5f8f16fd40c25d40314e6497b39375808aba186f48d
a42eefa895
2021-10-04T20:09:31.527537000+0200
2021-10-04T20:09:43.407678000+0200
2021-10-04T20:09:43.407911000+0200
2021-10-04T20:09:43.408701000+0200
```

The `tcp.payload` field contains hexadecimal data. Let's extract and reconstruct it:

```
tshark -r capture.flag.pcap -Y "tcp.port==9002" -T fields -e frame.time -e tcp.payload
| sort | cut -f2 -d '$'\t' | xxd -r -p
```

Output:

```
Salted__<K&???,J????o?
%??I{97?????.熙
```

This looks like the beginning of a DES3-encrypted file (the `Salted__` header is a known OpenSSL marker).

Step 6: Save the Encrypted File

Let's save this payload to a file named `file.des3` :

```
tshark -r capture.flag.pcap -Y "tcp.port==9002" -T fields -e frame.time -e tcp.payload  
| sort | cut -f2 -d $'\t' | xxd -r -p > file.des3
```

Step 7: Decrypt the File with OpenSSL

```
openssl des3 -d -salt -in file.des3 -out file.txt -k <REDACTED>
```

Output:

```
*** WARNING : deprecated key derivation used.  
Using -iter or -pbkdf2 would be better.
```

Note: The warning is normal. OpenSSL is alerting that the key derivation method is older, but the decryption still works.

Step 8: Read the Decrypted Flag

```
cat file.txt
```

Output:

```
picoCTF{<REDACTED>}
```

Alternative Method: Manual Packet Extraction

If you prefer to extract the encrypted file manually using Wireshark:

- 1. Filter packets with `tcp.port == 9002`
- 2. Look for packets containing **TCP payload** (non-zero length)
- 3. Right-click on a packet → **Follow** → **TCP Stream**
- 4. In the “Show data as” dropdown, select **Raw**
- 5. Save the raw data as `file.des3`
- 6. Decrypt with OpenSSL as shown above

Final Result

picoCTF{<REDACTED>}

Note: The actual flag is redacted here. In the real challenge, decrypting the file will reveal the complete flag.

Key Concepts

Concept	Description
PCAP Analysis	Network packet captures contain timestamped records of all traffic, useful for forensic investigations.
TCP Stream Following	Wireshark feature that reassembles and displays the complete conversation between two hosts.
Packet Filtering	Using display filters (<code>-Y</code> option in tshark) to focus on specific traffic patterns (e.g., <code>tcp.port==9002</code>).
Payload Extraction	Recovering data from TCP payloads using field extraction and hex conversion.

Concept	Description
Data Reconstruction	Combining multiple packets by sorting timestamps chronologically.
DES3 Encryption	Triple DES (3DES) encryption algorithm, commonly used with OpenSSL.
OpenSSL Salt	The <code>Salted__</code> header in encrypted files indicates that a salt was used during encryption.
tshark Advantages	Automated, scriptable packet analysis compared to manual GUI inspection.

Pro Tips

1. **Always follow TCP streams** – Chat conversations, file transfers, and command exchanges are often visible in TCP streams.
2. **Look for command strings** – Users often type commands in chats (like `openssl`), which can reveal encryption methods and passwords.
3. **Check unusual ports** – File transfers often occur on non-standard ports (like 9002 in this case).
4. **The `Salted__` header** – When you see this at the beginning of a file, it's almost certainly an OpenSSL-encrypted file (using `-salt` option).
5. **Sort by timestamp** – When extracting packet payloads, always sort by timestamp to reconstruct the data in the correct order.